Dr Alex Potanin

Associate Professor, School of Computing
College of Systems and Society, Australian National University
Adjunct Professor, S3D, Carnegie Mellon University
Senior Visiting Fellow, Trustworthy Systems, UNSW Sydney

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At a Glance

- **Track Record:** 100+ publications across two decades, 6 distinguished/best paper awards, 80+ students supervised at all levels, $3M+ in research funding
- **Research Areas:** Cyber Security (Capabilities, Language Security), Software Engineering, Programming Language Design & Implementation, Type Systems (Ownership, Immutability), Software Verification, Quantum Computing Verification, Embedded Systems & Edge AI
- **Leadership:** Chair of SPLASH Steering Committee (2024–2026), Member at Large and Treasurer of ACM SIGPLAN Executive Committee 2024–2027, General Chair of SPLASH/OOPSLA 2022, Research Committee Co-Chair of OOPSLA 2024, Program Committee Chair of APLAS 2025, Permanent Member of IFIP WG 2.4
- **Education:** BSc (Mathematics & Computer Science), BSc Hons, PhD — *Victoria University of Wellington, NZ*
- **Memberships:** Senior Member of the ACM, Member of Engineers Australia, Permanent Member of IFIP 2.4 Working Group on Software Implementation Technology

Research Impact

My ownership type system research directly influenced how the Rust programming language implements its ownership and lifetime system—one of Rust's defining features for memory safety. My foundational papers on generic ownership (OOPSLA 2006, 100+ citations) and generic immutability (FSE 2007, 100+ citations) showed how ownership and immutability guarantees can be incorporated "for free" into existing language designs. The "Performance of Open Source Applications" book cites my research that revolutionised performance evaluations in Talos and similar systems.

After a full-year sabbatical at Carnegie Mellon University, I co-created the Wyvern Programming Language with Professor Jonathan Aldrich—a novel general-purpose language designed with security and usability as primary goals. The CUE configuration language, used widely within Alibaba's cloud, based its module system on Wyvern's design. Scala drew on our Wyvern research both for its path-dependent types and type member mechanisms and, more recently, for its adoption of the capabilities-and-effects combination that Wyvern pioneered. Wyvern produced numerous publications including work on type-specific languages (ECOOP 2014 Distinguished Paper) and decidable typing for type members (POPL 2020).

My **cyber security** research spans capability-based systems, language-level security mechanisms, and vulnerability discovery. My work uncovered several security flaws in Java-based web servers (Jenkins, Tomcat, JBoss) and .NET equivalents, published at ECOOP 2017 (Distinguished Artifact Award) and supported by $50,000 USD Oracle Labs funding. Wyvern's capability-based module system provides authority-safe security guarantees, and my current research explores CHERI capabilities hardware security and secure-by-construction software development approaches.

Most recently, I have established a significant research programme in **quantum program verification**, applying formal methods expertise from classical software to the rapidly emerging quantum computing domain. This work has already produced publications at top venues: embedding quantum program verification into Dafny (OOPSLA 2025) and validating quantum state preparation programs (ESOP 2026, Distinguished Paper Award). I am supervising multiple students in this area across ANU and Iowa State University, and collaborating with Assistant Professor Liyi Li on quantum symbolic execution and distributed quantum computation. This represents a natural and impactful extension of my verification and language design expertise into one of computing's most important frontiers.

Selected Award-Winning & High-Impact Publications

1. Liyi Li, Anshu Sharma, Zoukarneini Difaizi Tagba, Sean Frett, and Alex Potanin. *Validating Quantum State Preparation Programs.* ESOP 2026. **ETAPS/ESOP 2026 Distinguished Paper Award.**
2. Amos Robinson, Alex Potanin. *Pipit on the Post: Proving Pre- and Post-Conditions of Reactive Systems.* ECOOP 2024. **ECOOP 2024 Distinguished Artifact Award.**
3. Tobias Runge, Alex Potanin, Thomas Thum, and Ina Schaefer. *Traits: Correctness-by-Construction for Free.* FORTE 2022. **FORTE 2022 Best Paper Award.**
4. Jens Dietrich, Kamil Jezek, Shawn Rasheed, Amjed Tahir, Alex Potanin. *EvilPickles: DoS Attacks Based on Object-Graph Engineering.* ECOOP 2017. **ECOOP 2017 Distinguished Artifact Award.**
5. Cyrus Omar, Darya Kurilova, Ligia Nistor, Benjamin Chung, Alex Potanin, and Jonathan Aldrich. *Safely Composable Type-Specific Languages.* ECOOP 2014. **ECOOP 2014 Distinguished Paper Award.**
6. Yoav Zibin, Alex Potanin, Mahmood Ali, Shay Artzi, Adam Kiezun, and Michael D. Ernst. *Object and Reference Immutability using Java Generics.* FSE 2007. **ESEC/FSE 2007 ACM SIGSOFT Distinguished Paper Award.** 100+ citations.
7. Alex Potanin, James Noble, Dave Clarke, and Robert Biddle. *Generic Ownership for Generic Java.* OOPSLA 2006. 100+ citations. Directly influenced the Rust programming language.
8. Julian Mackay, Alex Potanin, Jonathan Aldrich, and Lindsay Groves. *Decidable Subtyping for Path Dependent Types.* POPL 2020. Reusable Artifact badge.
9. Feifei Cheng, Sushen Vangeepuram, Henry Allard, Seyed M.R. Jafari, Alex Potanin, and Liyi Li. *Embedding Quantum Program Verification into Dafny.* OOPSLA 2025.
10. Alex Potanin, Johan Ostlund, Yoav Zibin, Michael D. Ernst. *Immutability.* Book chapter in Aliasing in Object-Oriented Programming, LNCS 7850, Springer, 2013.
11. Alex Potanin, James Noble, Marcus Frean, and Robert Biddle. *Scale-free Geometry in Object-Oriented Programs.* Communications of the ACM, 2005. 300+ citations.

Research Funding

Period	Funder	Grant	Amount
2027	Department of Defence (Australia)	Automated, Compositional and AI-Assisted Verification of Embedded and Cyber-Physical Systems with Fiducia	$100,000
2026 – 2030	NSF (USA)	GOALI: SHF — Automated Verification and Validation for Quantum Software (US$850,000; co-PI; PI Liyi Li, Iowa State)	$1,200,000
2026 – 2027	Anthropic	Claude API Credits for Teaching at SOCO (US$355,000; shared with Dr Ben Swift)	$500,000 (in-kind)
2026	ANU CSS	Tier 2 Learning & Teaching Grant: "Badges as a Complementary Performance Indicator" (shared with B. Pereira Nunes, A. Chen, B. Swift, B. Bergin, C. Martin, D. Guo, D. Campbell, A. Martin, P. Höfner, R. Shome, F. Muehlboeck, S. Okai-Ugbaje)	$5,000
2025 – 2026	Skykraft	Research Consulting	$30,000
2024	DVCRI	University Strategic Fund towards Centre of Excellence on Trustworthy Systems	$36,000
2022	ANU	Startup Package Funding	$500,000
2021 – 2023	SHEADI	Faculty Strategic Initiative PhD Scholarship	$100,000
2020 – 2021	Robonomics Network	Research Grant (Nitrate Sensors)	$72,000
2019	Kyoto University	Visiting Professor Scholarship	$20,000
2017 – 2018	Oracle Corporation	Research Grant (Language Security)	$70,000
2013	Carnegie Mellon University	Visiting Researcher Funding (DARPA Lablet)	$25,000
2012	Mozilla Foundation	Research Grant	$15,000
2009 – 2011	Royal Society of NZ	Marsden Fast Start Grant	$300,000
2009	RSNZ	ISAT Grant	$5,420
2007 – 2016	VUW	University Research Fund (multiple rounds)	$47,000
		Total	**$3,025,420**

Students and Mentoring

Over 80 students supervised across two decades at ANU, VUW, CMU, KIT, and Iowa State. My former students and collaborators have gone on to leading positions:

- Julian Mackay — Proof Engineer, Kry10; formerly Lecturer at Victoria University of Wellington and MBIE Science Whitinga Fellow
- Radu Muschevici — Assistant Professor, University of Nottingham Malaysia
- Amos Robinson — Computer Scientist, Microsoft
- Justin Lubin — PhD candidate, UC Berkeley
- Darya Melicher — Product Manager, SF Bay Area (previously Google)
- Paley Li — Postdoc, Czech Technical University (with Jan Vitek)
- Hannes Mehnert — Co-founder, Robur cooperative
- Felix Shi — Senior Security Architect, Xero
- Garming Sam — Security Platform Engineer, Westpac NZ
- Constantine Dymnikov — Solutions Architect, NZ Ministry of Justice

Current Students (12)

Student	Level	Topic
Yan Liu*	Postdoc (ANU)	Fiducia (Research Fellow), 2025–2027
Sasha Pak*	PhD (ANU)	Partial Borrows in Rust Verified with Mechanised Aeneas (with Ilya Sergey, Fabian Mühlböck)
Abolfazl Sharifi*	PhD (ANU)	Verification of Concurrent Data Structures in Rust using Iris
Edwin Singh*	PhD (ANU, PT)	Why Language Designers Do Modules The Way They Do
Billy Jaffray*	Honours (ANU)	Verifying Lock Free Data Structures using Verus
Charlotte Fulham*	Honours (ANU)	Verified Parser in Rust for Fiducia, 2026 S2 – 2027 S1
Haoyu Wu	PhD (ANU)	Nomnom Type Checking (with Fabian Mühlböck)
Carlo Zancanaro	PhD (ANU)	Gradual Typing and Type Polymorphism
Julia Groß	PhD (ANU, PT)	Energy Trading in Blockchains (with Jennifer Ferreira, Ramesh Rayudu)
Maximillian Kodetzki	PhD (KIT)	X-by-Construction (with Ina Schaefer)
Jakob Jerebica	PhD (KIT)	Formalisation of Pancake by Construction with Real World Driver Examples (with Ina Schaefer, Michael Norrish)
Alyssa Herald	Honours (ANU)	Security of Capabilities in CHERI

* Primary supervisor.

Selected Completed Students

- Charlotte Fulham (RA, ANU, June–September 2026) — Fiducia Microkit Backend (with Yan Liu)
- N Shalini Srinivasan (RA, ANU, June–August 2026) — Fiducia Model Checking (with Yan Liu)
- David Young (PhD, Kansas, PT, 2023–2026) — PCM's for Abstracted Layouts across Multiple Fields: from Heaps to Quantum
- Jack Hackshaw (Honours, ANU, 2025–2026) — Pancake in Loom in Lean
- Rishita Sarkar (BAC R&D, ANU, 2025–2026) — Quantum Computing Testing
- Libby Boas (PhB ASC, ANU, 2025–2026) — Quantum Computing Distributed Computation
- Charlotte Fulham (Eng ASC, ANU, 2026 S1) — Fiducia to Microkit Bridge Implementation
- Zara Hassan (PhD, ANU, 2023–2025, graduated 2026) — Reproducibility Debt in Scientific Software (with Christoph Treude, Michael Norrish, Graham Williams) → working with the Australian Research Data Commons
- Fahimeh Hoseinnia (PhD, VUW, 2023–2026) — Agricultural Pollution Data Marketplace and DAOs (with Jennifer Ferreira, James Quilty) → Data Analyst, Arthur D Riley & Co, NZ
- Iko-Ojo Simon (PhD, ANU, 2023–2025) — Algorithmic Debt → Postdoc, University of São Paulo, Brazil

- Amos Robinson (Postdoc, ANU, 2023–2025) — Higher Level Invariants with Lark → Microsoft
- Tobias Runge (PhD, KIT, 2019–2022) — CbC with Ownership and Traits (with Ina Schaefer)
- Julian Mackay (PhD, VUW, 2015–2019) — Wyvern Type Members (with Jonathan Aldrich) → Lecturer, VUW → Proof Engineer, Kry10
- Darya Melicher (PhD, CMU, 2013–2019) — Wyvern Modules (with Jonathan Aldrich) → Google → Product Manager, SF Bay Area
- Ahmed Khalifa (PhD, VUW, 2010–2013) — Ownership and Immutability in the Real World
- Garming Sam (BE Hons, VUW, 2015) — Refactorings in Rust → Westpac NZ
- Radu Muschevici (MSc, VUW, 2007–2008) — Multimethods → Asst. Prof., U. Nottingham Malaysia
- Constantine Dymnikov (MSc, VUW, 2011) — Modular Ownership Inference → NZ MoJ
- Paley Li (BSc Hons, VUW, 2008) — Ownership and Immutability → Postdoc, Czech Technical U.

As Associate Director of HDR at ANU School of Computing (2023–2025), I managed admissions, supervision, and progress for over 200 PhD students, created a monitoring app to proactively track student progress, and revamped the admissions and scholarship process.

Employment Record

2022 – Present	Associate Professor, Australian National University
2023 – Present	Adjunct Professor, S3D, Carnegie Mellon University, USA
2026 – Present	Senior Visiting Fellow, Trustworthy Systems, School of Computer Science and Engineering, UNSW Sydney
2025	Research Consultant, Usability Dynamics Inc, NC, USA
2025	Research Consultant, Skykraft, Canberra, Australia
2023 – 2025	Associate Director, HDR, School of Computing, ANU
2022 – 2025	Adjunct Professor, ECS, VUW, NZ
2022	Deputy Head of School, Victoria University of Wellington
2021 – 2022	Associate Dean (Students), Victoria University of Wellington
2018 – 2022	Associate Professor, Victoria University of Wellington
2019/20 (sabbatical)	Associate Professor, Kyoto University, Japan
2010 – 2017	Senior Lecturer, Victoria University of Wellington
2013 (sabbatical)	Research Scholar, Carnegie Mellon University
2006 – 2009	Lecturer, Victoria University of Wellington
2005 – 2006	Software Engineer, Innaworks Limited
2004	Visiting Researcher, Purdue University
2001 – 2003	Software Engineer, Catalyst Systems Limited (part-time)

Professional Memberships

2024 – 2027	Member at Large and Treasurer, *ACM SIGPLAN Executive Committee*
2024 – 2026	Chair, *SPLASH Steering Committee*
2023 – Present	Member of Engineers Australia
2002 – Present	Member of the ACM (currently: Senior Member)
2018 – 2022	Member of Engineering New Zealand
2010 – 2014	Conference Coordinator, CORE (CS Departments of Australia/NZ) Executive

Awards

- 2026 – ESOP/ETAPS 2026 Distinguished Paper Award
- 2026 – Innovation Award Finalist (2nd place, shared), ASCA Pitch Day 2026 (Resilient Command & Control), for *Provably Secure Tactical Edge C2 Platform* (with Gernot Heiser, UNSW)
- 2025 – ANU CSS Education Awards: Excellence in Supervision (Commendation)
- 2025 – ANU CSS Education Awards: Outstanding Contribution to Student Experience (Commendation)
- 2024 – ECOOP 2024 Distinguished Artifact Award
- 2022 – FORTE 2022 Best Paper Award
- 2017 – ECOOP 2017 Distinguished Artifact Award
- 2014 – ECOOP 2014 Distinguished Paper Award
- 2007 – ESEC/FSE ACM SIGSOFT Distinguished Paper Award
- 2005 – 2nd Prize and Judges Prize at ICFP Programming Contest 2005 and 2003
- 2004 – Claude McCarthy Fellowship; 2003 – J. L. Stewart Scholarship

- 2003 – 2nd Place, ACM International Research Competition (via OOPSLA 2002 SRC)
- 2002 – Freemasons Scholarship; 2001 – Datacom Scholarship in Computer Science
- 1997 – Finalist, George Soros International Mathematics Olympiad
- 1996 – Winner, Russian City Tournament in Mathematics

Community Leadership and Service

Conference Leadership

- Chair, SPLASH Steering Committee, 2024–2026
- Member at Large and Treasurer, ACM SIGPLAN Executive Committee, 2024–2027 — turned around the budget from a ∼$400K/year loss to a sustainable $100K–$200K/year surplus by synchronising conference registrations and expenses, developing clear policies, and coordinating across steering committees
- General Chair, SPLASH/OOPSLA 2022 (combined with APLAS 2022), Auckland, NZ
- Research Committee Co-Chair, OOPSLA 2024 (with Bor-Yuh Evan Chang)
- Program Committee Chair, APLAS 2025
- General Chair, APLAS 2018, Wellington, NZ
- Program Co-Chair, APSEC 2016 (with Gail Murphy)
- PC Co-Chair, CATS 2010 and 2011 (with Taso Viglas)
- General Co-Chair, ACSW 2009, Wellington, NZ
- General Chair, 15th PhD Workshop and Doctoral Symposium at ECOOP 2005
- Chair, NOOL Workshop 2015 and 2017
- Workshops Co-Chair, SPLASH 2017 and 2018
- Publications Chair, SPLASH 2015–2017; ICFP 2018
- Virtualisation Co-Chair, SPLASH 2020 and 2021
- SPLASH/OOPSLA Steering Committee Member from 2020 (Chair from 2024)
- SAPLING Steering Committee Member from 2019

Program Committee Member

OOPSLA (2027 RC, 2021 ERC, 2020 RC, 2019 RC, 2014, 2011 ERC), PLDI (2025 RC, 2019 SRC, 2015 ERC), ECOOP (2023, 2022, 2019, 2016), ESOP 2021, APLAS (2021, 2016), GPCE (2020, 2019), FormaliSE (2023, 2022), Programming RC (2023, 2022), IWACO (2024, 2021, 2011, 2007), FSE-IVR 2025, HotSoS 2019, ICCQ 2021, APSEC 2017, NOOL 2016, FTfJP 2016, FOOL 2014, Onward! Papers 2023, ACSC (2010–2017).

University Service

- Associate Director, HDR, ANU School of Computing (2023–2025) — managed 200+ PhD students
- Program Convenor, BE Software Engineering, ANU (2022–present)
- DBI&E Working Group Lead, ANU School of Computing (2023–present)
- Deputy Head of School, VUW (2022) — 15 direct academic reports
- Associate Dean (Students), Faculty of Engineering, VUW (2021–2022) — 2000 students
- Program Director (Software Engineering), VUW (2017–2018) — led SWEN/NWEN/CYBR majors
- Program Director (Science), VUW (2019) — led introduction of AI/ML postgraduate program
- Postgraduate Coordinator, VUW (2012–2016) — 100 PhD/Masters students
- Engineering NZ Accreditation Panel member (2018) — re-accredited U. Canterbury BE
- NZ Qualifications Authority representative (2018) — accredited new BIT degree

Research Themes

Cyber Security

Capability-based security, authority-safe module systems, and preventing command injection and other attacks through language design. EvilPickles (ECOOP 2017, Distinguished Artifact) uncovered several security flaws in Java/JBoss/Jenkins and .NET servers—work supported by $50,000 USD Oracle Labs funding. Research on CHERI capabilities hardware security (Honours student Alyssa Herald). Secure-by-construction software development through the Wyvern language's capability-based module system (ECOOP 2017, HotSoS 2018) and type-specific languages to fight injection attacks (HotSoS 2014).

Embedded Systems & Edge AI

Trustworthy systems security for embedded and space platforms, including consulting for Skykraft and other industry partners. Exploring secure Edge AI deployment on NVIDIA Jetson Orin.

Quantum Computing

A major new research direction applying formal methods to quantum computing. Developed the first embedding of quantum program verification into the Dafny verification framework (OOPSLA 2025), enabling developers to write and verify quantum programs using familiar classical tools. Our work on validating quantum state preparation programs (ESOP 2026, Distinguished Paper) addresses a critical challenge in ensuring correctness of quantum circuits. Projects in this area include quantum symbolic execution, quantum computing testing, and distributed quantum computation, in collaboration with Liyi Li (Iowa State).

The Wyvern Language

A secure general-purpose programming language using object capabilities and effects, co-developed with Jonathan Aldrich at CMU over 2013–2023. The CUE configuration language based its module system on Wyvern. Scala drew on our Wyvern research for its path-dependent types and type members, and more recently adopted the capabilities-and-effects combination that Wyvern pioneered. Produced 12+ co-authored papers (`http://wyvernlang.github.io/`).

Ownership & Immutability

Pioneered Generic Ownership, showing how type polymorphism provides ownership support. This approach was adopted by the Rust programming language as "lifetime parameters". Led the Marsden-funded project (2008–2012) unifying ownership and immutability in a single mathematical framework.

Software Verification

Proving pre- and post-conditions of reactive systems with the Pipit framework (ECOOP 2024, Distinguished Artifact), and correct-by-construction programming approaches including traits (FORTE 2022 Best Paper) and flexible CbC (LMCS 2023). Research Fellow Yan Liu is working on trustworthy systems verification.

Software Engineering

Empirical studies of software development practices, developer tools and productivity, API usability, and evidence-based approaches to improving software quality and developer experience. Supervised research on reproducibility debt in scientific software (JSS 2025, FSE-IVR 2024, SPLASH 2025 poster), algorithm debt in ML/DL systems (ESE 2025, ICSME 2024), and automated refactoring in Rust (ACSC 2017, SPLASH 2025 posters). Led the Software Engineering program at VUW (2017–2018) and currently convenes BE Software Engineering at ANU.

Collaborations

- Assistant Professor Liyi Li, Iowa State University, USA (quantum program verification, symbolic execution)
- Professor Jonathan Aldrich, CMU, USA (Wyvern, capabilities, type members)
- Professor Ina Schaefer, KIT, Germany (correct-by-construction, information flow)
- Professor Atsushi Igarashi, Kyoto University, Japan (ownership verification)
- Professor Ilya Sergey, NUS, Singapore (Rust, deductive synthesis)

Referees

- Professor Peter Müller, *ACM Fellow*, ETH Zurich, Switzerland
- Professor Steve Blackburn, *ACM Fellow*, ANU/Google DeepMind
- Professor Jeff Foster, *ACM Fellow*, Tufts University, USA
- Professor David Lo, *ACM Fellow*, Singapore Management University
- Professor Anders Møller, Aarhus University, Denmark
- Professor Sophia Drossopoulou, Imperial College London, UK
- Professor Jan Vitek, Northeastern University, USA

Teaching Highlights

- **COMP 2300 Computer Architecture (ANU, 2027 S1):** Convening this core computer architecture course.
- **COMP 2100/COMP 6442 Software Construction (ANU, 2026 S2):** Convening this core software construction course.
- **COMP 3500/4500/8715 TechLauncher (ANU, 2025; 2026 S1):** Redesigned capstone course for

400+ students with external clients. Reduced tutor budget by half while increasing satisfaction. *Achieved 100% overall student satisfaction (COMP 3500); lifted COMP 8715 agreement from 55% to 81%.*
- **COMP 2120 Software Engineering (ANU, 2022–2024):** Created modern group-project-oriented course where 50+ teams contribute pull requests to large open-source projects.
- **SWEN 325 Mobile Development (VUW, 2018):** Created brand-new course from scratch. *Overall teaching evaluation: 1.0 (best possible).*
- **SWEN 430 Compiler Engineering (VUW, 2020):** Close to 1.0 teaching evaluations despite COVID year.
- **COMP 361 Algorithms (VUW, 2014):** Reinvented struggling course, increasing enrolments tenfold from 5–6 to ∼60.
- **SWEN 302 Agile Methods (VUW, 2014):** Rescued course from 4.0 evaluation (worst in ECS history) to 2.0 via complete redesign with time-boxing and internal clients.
- **AI/ML Program (VUW, 2019):** Led introduction of postgraduate AI and Machine Learning program as Program Director (Science).
- **Software Engineering Review (VUW, 2017):** Led first major overhaul of BE Software Engineering across 20+ staff, approved by Academic Board.

Consistent "overall effectiveness" scores around 1.6 at VUW (scale 1–5, 1 = outstanding). Perfect 1.0 scores for "Attitude towards students" across all years. Taught classes from 5 to 500 students across all levels.

Full Publications List

Book Chapters

1. James Noble, Alex Potanin, Toby Murray, Mark S. Miller. *Abstract and Concrete Data Types vs Object Capabilities.* In P. Müller and Ina Schaefer (Eds.): Principled Software Development. Springer, 2018.
2. Alex Potanin, Johan Ostlund, Yoav Zibin, Michael D. Ernst. *Immutability.* In D. Clarke et al. (Eds.): Aliasing in Object-Oriented Programming, LNCS 7850, pp. 233–269. Springer, 2013.

Journal Papers

3. Zara Hassan, Christoph Treude, Michael Norrish, Graham Williams, and Alex Potanin. *Reproducibility Debt: Insights from Practitioners — A Mixed-Methods Study of Causes, Effects, and Mitigation Strategies.* Empirical Software Engineering. Springer. Accepted for publication in September 2026.
4. Emmanuel Iko-Ojo Simon, Chirath Hettiarachchi, Fatemeh Fard, Alex Potanin, Hanna Suominen. *A Survey of Algorithm Debt in Machine and Deep Learning Systems: Definition, Smells, and Future Work.* ACM Computing Surveys. Volume 58, Issue 11, Article No. 296, pp. 1–35. Published 27 April 2026. doi:10.1145/3806391.
5. Emmanuel Iko-Ojo Simon, Chirath Hettiarachchi, Alex Potanin, Hanna Suominen, Fatemeh Fard. *Automated Detection of Algorithm Debt in Deep Learning Frameworks: An Empirical Study.* Empirical Software Engineering. Springer, 2026. Volume 31, Article 66 (Open Access). doi:10.1007/s10664-026-10807-5.
6. Zara Hassan, Christoph Treude, Michael Norrish, Graham Williams, and Alex Potanin. *Characterising Reproducibility Debt in Scientific Software: A Systematic Literature Review.* The Journal of Systems & Software. Volume 222, April 2025, 112327.
7. Tobias Runge, Tabea Bordis, Alex Potanin, Thomas Thum, Ina Schaefer. *Flexible Correct-by-Construction Programming.* Logical Methods in Computer Science. Volume 19, Issue 2, pp. 16:1–16:36. 2023.
8. Tobias Runge, Marco Servetto, Alex Potanin, Ina Schaefer. *Immutability and Encapsulation for Sound OO Information Flow Control.* ACM TOPLAS. Volume 45, Issue 1, Article No. 3 pp 1–35. 2022.
9. Isaac Oscar Gariano, Marco Servetto, Alex Potanin. *Using Capabilities for Strict Runtime Invariant Checking.* Science of Computer Programming, Volume 224, 2022.
10. Darya Melicher, Anlun Xu, Valerie Zhao, Alex Potanin, Jonathan Aldrich. *Bounded Abstract Effects.* ACM TOPLAS, Volume 44, Issue 1, March 2022.
11. Isaac Oscar Gariano, Marco Servetto, Alex Potanin, Hrshikesh Arora. *Iteratively Composing Statically Verified Traits.* EPTCS. Issue 299, Paper 7. 2019.
12. Chris Male, David Pearce, Alex Potanin, and Constantine Dymnikov. *Formalisation and Implementation of an Algorithm for Bytecode Verification of @NonNull Types.* Science of Computer Programming. Volume 76, Issue 7, pp. 587–608, 2011.
13. Alex Potanin, James Noble, Dave Clarke, and Robert Biddle. *Featherweight Generic Confinement.* Journal of Functional Programming. Volume 16, Number 6, pp. 793–811, 2006.
14. Alex Potanin, James Noble, Marcus Frean, and Robert Biddle. *Scale-free Geometry in Object-Oriented Programs.* Communications of the ACM. Pages 99–103. May 2005.

15. Alex Potanin, James Noble, and Robert Biddle. *Checking Ownership and Confinement*. Concurrency and Computation: Practice and Experience. Volume 16, Issue 7, pp. 671–687, 2004.

Refereed Conference Papers

16. Dariy Guzairov, Alex Potanin, Stephen Kell, and Alwen Tiu. *A Security Analysis of CheriBSD and Morello Linux*. ESORICS 2026.
17. Liyi Li, Anshu Sharma, Zoukarneini Difaizi Tagba, Sean Frett, and Alex Potanin. *Validating Quantum State Preparation Programs*. ESOP 2026. **Distinguished Paper.**
18. Feifei Cheng, Sushen Vangeepuram, Henry Allard, Seyed M.R. Jafari, Alex Potanin, and Liyi Li. *Embedding Quantum Program Verification into Dafny*. OOPSLA 2025.
19. Maximilian Kodetzki, Tabea Bordis, Alex Potanin, Ina Schaefer. *X-by-Construction: Towards Ensuring Non-Functional Properties in by-Construction Engineering*. Onward! 2025.
20. Emmanuel Iko-Ojo Simon, Chirath Hettiarachchi, Alex Potanin, Hanna Suominen, and Fatemeh Fard. *Automated Detection of Algorithm Debt in Deep Learning Frameworks*. ICSME 2024.
21. David Young, Ziyi Yang, Ilya Sergey, Alex Potanin. *Higher-Order Specifications for Deductive Synthesis of Programs with Pointers*. ECOOP 2024.
22. Amos Robinson, Alex Potanin. *Pipit on the Post: Proving Pre- and Post-Conditions of Reactive Systems*. ECOOP 2024. **Distinguished Artifact.**
23. Zara Hassan, Christoph Treude, Michael Norrish, Graham Williams, Alex Potanin. *Reproducibility Debt: Challenges and Future Pathways*. FSE-IVR 2024.
24. Tobias Runge, Alexander Kittelmann, Marco Servetto, Alex Potanin, and Ina Schaefer. *Information Flow Control-by-Construction for an Object-Oriented Language*. SEFM 2022.
25. Tobias Runge, Alex Potanin, Thomas Thum, and Ina Schaefer. *Traits: Correctness-by-Construction for Free*. FORTE 2022. **Best Paper.**
26. Manish Singh, Lindsay Groves, and Alex Potanin. *A Relaxed Balanced Lock-free Binary Tree*. PDCAT 2020.
27. Julian Mackay, Alex Potanin, Jonathan Aldrich, and Lindsay Groves. *Syntactically Restricting Bounded Polymorphism for Decidable Subtyping*. APLAS 2020.
28. Julian Mackay, Alex Potanin, Jonathan Aldrich, and Lindsay Groves. *Decidable Subtyping for Path Dependent Types*. POPL 2020. Reusable Artifact badge.
29. Aaron Craig, Alex Potanin, Lindsay Groves and Jonathan Aldrich. *Capabilities: Effects for Free*. ICFEM 2018. Pp 231–247. Springer.
30. Jens Dietrich, Kamil Jezek, Shawn Rasheed, Amjed Tahir, Alex Potanin. *EvilPickles: DoS Attacks Based on Object-Graph Engineering*. ECOOP 2017. **Distinguished Artifact.**
31. Darya Melicher, Yangqingwei Shi, Alex Potanin, Jonathan Aldrich. *A Capability-Based Module System for Authority Control*. ECOOP 2017.
32. Garming Sam, Nicholas Cameron and Alex Potanin. *Automated Refactoring of Rust Programs*. ACSC 2017.
33. Joseph Lee, Jonathan Aldrich, Troy Shaw, Alex Potanin. *A Theory of Tagged Objects*. ECOOP 2015.
34. Cyrus Omar, Darya Kurilova, Ligia Nistor, Benjamin Chung, Alex Potanin, and Jonathan Aldrich. *Safely Composable Type-Specific Languages*. ECOOP 2014. **Distinguished Paper.**
35. Marco Servetto, Julian Mackay, Alex Potanin, and James Noble. *The Billion-Dollar Fix: Safe Modular Circular Initialisation with Placeholders and Placeholder Types*. ECOOP 2013. Pp 205–229.
36. Constantine Dymnikov, David Pearce and Alex Potanin. *OwnKit: Inferring Modularly Checkable Ownership Annotations for Java*. ASWEC 2013. Pp 181–190.
37. Alex Potanin, Monique Damitio and James Noble. *Are Your Incoming Aliases Really Necessary? Counting the Cost of Object Ownership*. ICSE 2013. Pp 742–751.
38. Daniel Atkins, Alex Potanin and Lindsay Groves. *The Design and Implementation of Clocked Variables in X10*. ACSC 2013.
39. Jan Larres, Alex Potanin and Yuichi Hirose. *A Study of Performance Variations in the Mozilla Firefox Web Browser*. ACSC 2013.
40. Hien Tran, Craig Anslow, Stuart Marshall, Alex Potanin, Mairead de Roiste. *Lessons Learnt from Collaboratively Creating Maps on a Touch Table*. CHINZ 2011.
41. Yoav Zibin, Alex Potanin, Paley Li, Mahmood Ali, Michael D. Ernst. *Ownership and Immutability in Generic Java*. OOPSLA 2010. Pp. 598–617.
42. Radu Muschevici, Alex Potanin, Ewan Tempero, and James Noble. *Multiple Dispatch in Practice*. OOPSLA 2008. Pp. 563–582.
43. Chris Male, David Pearce, Alex Potanin, and Constantine Dymnikov. *Java Bytecode Verification for @NonNull Types*. CC 2008. Pp 229–244.

44. Neil Ramsay, Stuart Marshall, and Alex Potanin. *Annotating UI Architecture with Actual Use.* AUIC 2008.
45. Yoav Zibin, Alex Potanin, Mahmood Ali, Shay Artzi, Adam Kiezun, and Michael D. Ernst. *Object and Reference Immutability using Java Generics.* FSE 2007. **ESEC/FSE 2007 Distinguished Paper.**
46. Alex Potanin, James Noble, Dave Clarke, and Robert Biddle. *Generic Ownership for Generic Java.* OOPSLA 2006. Pp. 311–324.
47. Alex Potanin, James Noble, and Robert Biddle. *Snapshot Query-Based Debugging.* ASWEC 2004. Pp 251–261.

Refereed Workshop Papers

48. Zara Hassan, Christoph Treude, Graham Williams, Michael Norrish, Alex Potanin. *Managing Reproducibility Debt in Scientific Software: A Practical Framework.* SERS 2026.
49. Sasha Pak, Fabian Muehlboeck, Alex Potanin. *A Verified Thread-Safe Array in Rust.* IWACO 2025.
50. Abhaas Goyal, Alex Potanin and Jonathan Aldrich. *A Comparative Study of Traditional versus Capability-Based Module Systems for Modern Programming Languages.* PLATEAU 2024.
51. Amos Robinson and Alex Potanin. *Pipit: Reactive Systems in F Star (Extended Abstract).* TyDe 2023.
52. Baptiste Pauget, David Pearce, and Alex Potanin. *Towards Compilation of an Imperative Language for FPGAs.* VMIL 2018.
53. James Noble, Sophia Drossopoulou, Mark S Miller, Toby Murray and Alex Potanin. *Abstract Data Types in Object-Capability Systems.* IWACO 2016.
54. Du Li, Alex Potanin, and Jonathan Aldrich. *Delegation vs Inheritance for Typestate Analysis.* FTfJP 2015.
55. Darya Kurilova, Alex Potanin, and Jonathan Aldrich. *Wyvern: Impacting Software Security via Programming Language Design.* PLATEAU 2014.
56. James Noble and Alex Potanin. *On Owners-as-Accessors.* IWACO 2014.
57. Jonathan Aldrich, Cyrus Omar, Alex Potanin and Du Li. *Language-Based Architectural Control.* IWACO 2014.
58. Cyrus Omar, Benjamin Chung, Darya Kurilova, Alex Potanin and Jonathan Aldrich. *Type-Directed, Whitespace-Delimited Parsing for Embedded DSLs.* GlobalDSL 2013.
59. Ligia Nistor, Darya Kurilova, Stephanie Balzer, Benjamin Chung, Alex Potanin and Jonathan Aldrich. *Wyvern: A Simple, Typed, and Pure Object-Oriented Language.* MASPEGHI 2013.
60. Marco Servetto, David Pearce, Lindsay Groves, and Alex Potanin. *Balloon Types for Safe Parallelisation over Arbitrary Object Graphs.* WoDeT 2013.
61. Daniel Atkins, Alex Potanin, Lindsay Groves. *Clocked References in X10.* LaME 2012.
62. Julian Mackay, Hannes Mehnert, Alex Potanin, Lindsay Groves, Nicholas Cameron. *Encoding Featherweight Java with Assignment and Immutability using The Coq Proof Assistant.* FTfJP 2012.
63. Yoav Zibin, Alex Potanin, Paley Li, Mahmood Ali, Michael D. Ernst. *Ownership and Immutability in Generic Java (OIGJ).* PLDE 2010.
64. Paley Li, Stephen Nelson, and Alex Potanin. *Ownership for Relationships.* IWACO 2009.
65. Paley Li, Alex Potanin, James Noble, and Lindsay Groves. *Towards Unifying Immutability and Ownership.* IWACO at ECOOP 2008.
66. Christo Fogelberg, Alex Potanin, and James Noble. *Ownership Meets Java.* IWACO at ECOOP 2007.
67. Alex Potanin, James Noble, Tian Zhao, Jan Vitek. *A High Integrity Profile for Memory Safe Programming in Real-time Java.* JTRES 2005.
68. Alex Potanin, James Noble, Dave Clarke, Robert Biddle. *Featherweight Generic Ownership.* FTfJP at ECOOP 2005.
69. Alex Potanin, James Noble, Dave Clarke, Robert Biddle. *Defaulting Generic Java to Ownership.* FTfJP at ECOOP 2004.
70. Alex Potanin, James Noble, Dave Clarke, and Robert Biddle. *Featherweight Generic Confinement.* FOOL at POPL 2004.
71. James Noble, Robert Biddle, Ewan Tempero, Alex Potanin, and Dave Clarke. *Towards a Model of Encapsulation.* IWACO at ECOOP 2003.
72. Alex Potanin and James Noble. *Checking Ownership and Confinement Properties.* FTfJP at ECOOP 2002.

Edited Journals

73. Alex Potanin and Bor-Yuh Evan Chang (Associate Editors). *OOPSLA 2024 PACMPL Volume 8 Issue OOPSLA1 and OOPSLA2.*
74. Alex Potanin and Gail Murphy (Editors). *Special Issue on APSEC 2016.* Science of Computer Pro-

gramming. Volume 163, October 2018.

75. Alex Potanin (Editor). *Special Issue on NOOL 2015.* Journal of Object Technology. Volume 16, no. 2, April 2017.
76. Taso Viglas and Alex Potanin (Editors). *Special Issue on CATS 2011.* IJFCS, 2013. Vol. 24, Issue 1.
77. Taso Viglas and Alex Potanin (Editors). *Special Issue on CATS 2010.* CJTCS, May 2011.

Patent

- Stephen Ming Ko Cheng, Alex Potanin, Christopher Michael Andreae, Simon Marsh David Robinson. A computer implemented translation method. EP2122464A4.

Other Refereed Publications

78. Yi-Chieh Lee, Nattapat Boonprakong, Yugin Tan, Harold Soh, Alex Potanin, Viraj Kumar, Anoop K. Sinha, Chen Qian, Paul Denny, Mennatallah El-Assady, et al. *Reshaping Undergraduate Computer Science Education in the Generative AI Era.* Workshop report, arXiv:2606.07545, 2026.
79. Asma Shakil, Marie Devlin, KellyAnn Fitzpatrick, Sarah Carruthers, Mirela Gutica, Ronnie Howard, Stoney Jackson, Chris Johnson, Edward Latorre, Tyler Menezes, Joseph Mertz, Tominiyi Olupitan, Alex Potanin, Floris Westerman. *Exploring the Impact of Gen-AI on Team-Based Computing Capstone Projects.* ITiCSE 2026 Working Group, Madrid, Spain. ACM.
80. Sasha Pak, Richard Willie, Umang Mathur, Fabian Muehlboeck, Alex Potanin. *View Types in Rust.* Poster at SPLASH 2025.
81. Matthew Britton, Alex Potanin, Sasha Pak. *Verifying Extract Method Refactoring in Rust.* Poster at SPLASH 2025.
82. Zara Hassan, Christoph Treude, Graham Williams, Michael Norrish, Alex Potanin. *Reproducibility Debt in Scientific Software.* Poster at SPLASH 2025.
83. Manish Singh, Lindsay Groves, Alex Potanin. *A Relaxed Balanced Non-Blocking Binary Search Tree.* Poster, ICPP 2019.
84. Isaac Oscar Gariano, Marco Servetto, Alex Potanin, Hrshikesh Arora. *Iteratively Composing Statically Verified Traits.* Extended Abstract, VPT 2019.
85. Darya Melicher, Yangqingwei Shi, Valerie Zhao, Alex Potanin, Jonathan Aldrich. *Using Object Capabilities and Effects to Build an Authority-Safe Module System.* Poster, HotSoS 2018.
86. Aaron Craig, Alex Potanin, Lindsay Groves, Jonathan Aldrich. *Capabilities and Effects.* OCAP 2017.
87. Darya Melicher, Yangqingwei Shi, Valerie Zhao, Alex Potanin, Jonathan Aldrich. *Using Object Capabilities and Effects to Build an Authority-Safe Module System.* OCAP 2017.
88. Jonathan Aldrich and Alex Potanin. *Usably Expressing and Enforcing Design in Wyvern.* NOOL 2017.
89. Jens Dietrich, Kamil Jezek, Shawn Rasheed, Amjed Tahir, Alex Potanin. *EvilPickles (Artifact).* DARTS.
90. Darya Melicher, Yangqingwei Shi, Jonathan Aldrich. *A Capability-Based Module System for Authority Control (Artifact).* DARTS.
91. Jonathan Aldrich and Alex Potanin. *Delegation Revisited.* NOOL 2016.
92. Jonathan Aldrich and Alex Potanin. *Naturally Embedded DSLs.* DSLDI 2016.
93. Darya Kurilova, Alex Potanin, and Jonathan Aldrich. *Modules in Wyvern: Advanced Control over Security and Privacy.* Poster, HotSoS 2016.
94. Joseph Lee, Jonathan Aldrich, Troy Shaw, and Alex Potanin. *A Theory of Tagged Objects (Artifact).* DARTS 2015.
95. Cyrus Omar et al. *Safely Composable Type-Specific Languages.* Poster, ECOOP 2014.
96. Darya Kurilova et al. *Type-Specific Languages to Fight Injection Attacks.* Poster, HotSOS 2014.
97. Cyrus Omar et al. *Extensible Type-Driven Parsing for Embedded DSLs in Wyvern.* Parsing@SLE 2013.
98. Cyrus Omar et al. *Extensible Type-Driven Parsing for Embedded DSLs in Wyvern.* Poster, SPLASH 2013.
99. Jonathan Aldrich et al. *DSL support in Wyvern Language.* DSLDI 2013.
100. Jan Larres, Alex Potanin, and Yuichi Hirose. *Performance Variance Evaluation on Mozilla Firefox.* NZCSRSC 2011.
101. Mairead de Roiste, Hien Tran, and Alex Potanin. *What makes a map?* IRLOGI 2010.
102. Chris Andreae, Donald Gordon, Alex Potanin, James Noble, Robert Biddle. *Terrier: Static Query-Based Debugging in Eclipse.* Poster, OOPSLA 2004.
103. Alex Potanin. *Generic Ownership: Practical Ownership Control in Programming Languages.* Doctoral Symposium, OOPSLA 2004.
104. Alex Potanin. *Practical Ownership Control in Programming Languages.* Doctoral Symposium, ECOOP 2004.
105. Alex Potanin. *A Tool for Ownership and Confinement Analysis of the Java Object Graph.* Poster and

SRC entry, OOPSLA 2002. 2nd place in graduate division; accepted to ACM SRC Grand Finals; 2nd place in undergraduate category.

Theses

- Alex Potanin. *Generic Ownership — A Practical Approach to Ownership and Confinement in OO Programming Languages*. PhD thesis, 2007. Supervised by James Noble, Dave Clarke, and Robert Biddle.
- Alex Potanin. *The Fox — A Tool for Object Graph Analysis*. Honours report, First Class Honours, Victoria University of Wellington, 2002.